

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Audio Upload	Allows users to upload voice recordings for analysis.	Streamlit UI	Done	High
2	Speech-to-Text Conversion	Converts uploaded speech into text using Whisper AI.	Whisper Module	Done	High
3	Concept Understanding Analysis	Evaluates the conceptual correctness of the spoken explanation.	NLP Analysis	Done	High
4	Confidence Score Generation	Calculates the confidence level of the explanation.	Analysis Engine	Done	Medium
5	Audio Waveform Visualization	Displays the waveform of the uploaded audio.	Librosa + Matplotlib	Done	Medium
6	PDF Report Generation	Generates a downloadable report containing evaluation results.	ReportLab	Done	High
7	User-Friendly	Interactive interface for	Streamlit	Done	Medium

	Dashboard	uploading audio and viewing results.			
8	Error Handling	Handles invalid audio files and processing errors gracefully.	Validation Module	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Speech-to-Text, Concept Analysis, PDF Report, Audio Upload, User Dashboard

Additional / Nice-to-Have Features:

- Audio waveform visualization
- Downloadable PDF report
- Interactive Streamlit dashboard
- User-friendly interface

Features Tested & Verified:

- Audio file upload
- Speech-to-text conversion
- Concept analysis accuracy
- Confidence score generation
- PDF report generation
- Audio waveform visualization
- Error handling for invalid files

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Audio Upload, Interactive Dashboard
2	Backend / Logic	4	Speech-to-Text, Concept Analysis, Confidence Score, Error Handling

3	Database / Storage	0	Not Applicable (No database used)
4	API / Integration	2	Whisper AI, ReportLab Integration
5	Security / Authentication	0	Not Implemented